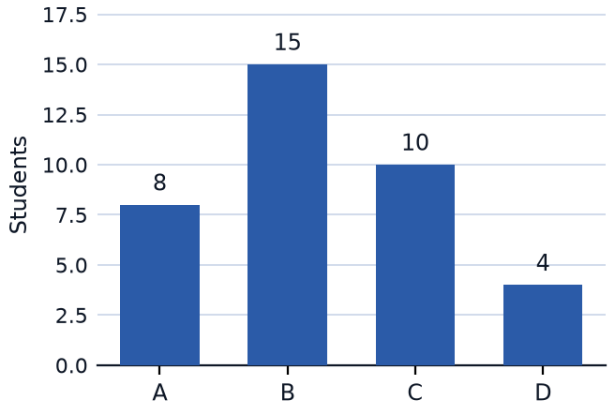
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

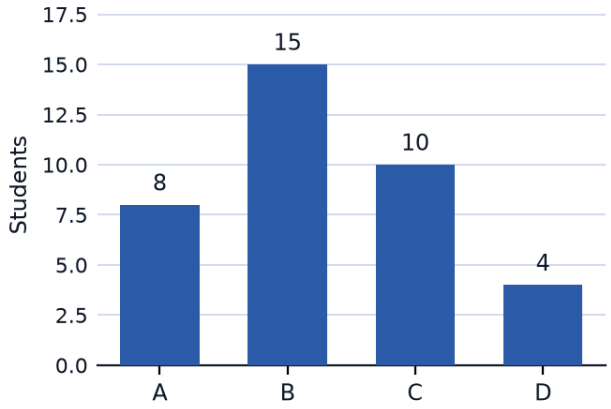
1 **[Foundational]** • **Target: Data representation**

The bar chart shows the distribution of grades in a class of 37 students. Find the probability that a randomly selected student earned a grade of A or B, as a percent to the nearest whole percent.



2 **[Foundational]** • **Target: Data representation**

Using the same bar chart, find the probability that a randomly selected student scored a C or D grade, as a percent to the nearest whole percent.



3 **[Foundational]** • **Target: Permutations**

How many ways can 5 distinct books be arranged on a shelf?

4 **[Foundational]** • **Target: Permutations**

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?

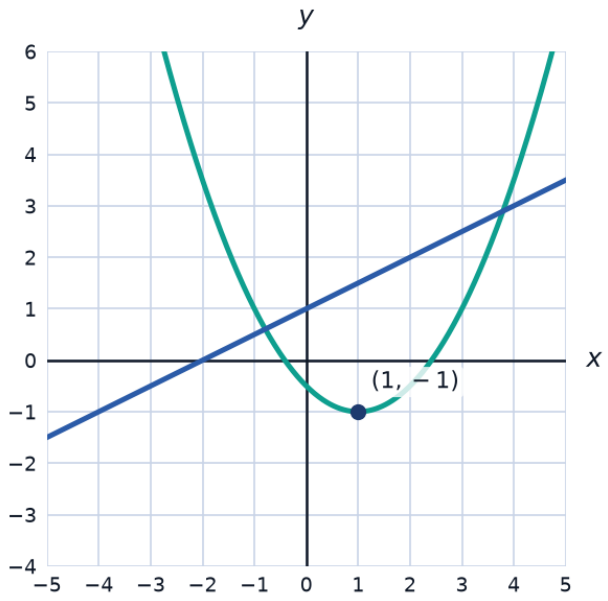
5 **[Foundational]** • **Target: Permutations**

How many distinct arrangements are there of the letters in the word APPLE?

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6 [Foundational] • Target: Combinations

How many ways can a committee of 4 be chosen from 11 people?



7 [Foundational] • Target: Combinations

A pizza shop offers 8 toppings. How many different 3-topping pizzas can be made?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Combinations

From a group of 6 men and 5 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?

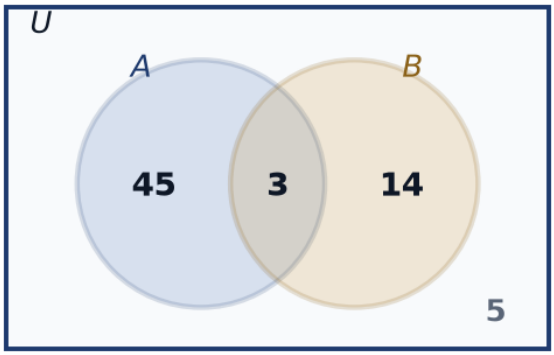
9 [Intermediate] • Target: Probability rules

A fair die is rolled. Find $P(\text{even or greater than } 4)$.

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10 [Intermediate] • Target: Probability rules

$P(A) = 0.45$ and $P(B) = 0.3$, and A and B are mutually exclusive. Find $P(A \text{ or } B)$.



11 [Intermediate] • Target: Probability rules

A card is drawn from a standard 52-card deck. Find $P(\text{not a face card})$.

12 [Intermediate] • Target: Conditional probability and independence

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?

13 [Intermediate] • Target: Conditional probability and independence

In a class, 60% of students play sports, and 25% of all students both play sports and are honor students. Find $P(\text{honor student} \mid \text{plays sports})$.

14 [Intermediate] • Target: Conditional probability and independence

Events C and D are independent, with $P(C) = 0.7$ and $P(D) = 0.6$. Find $P(C \cup D)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

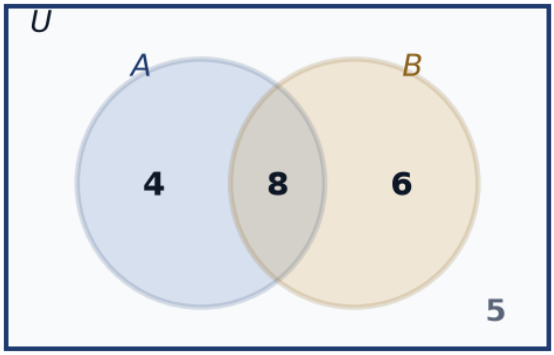
15 [Challenging] • Target: Measures of center and spread • Cross-Lesson: 1.01 Advanced Algebra

Find the mean and median of the data set $12, 15, 18, 20, 25$.

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16 [Challenging] • Target: Measures of center and spread • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Find the population standard deviation of the data set $4, 8, 6, 5, 3$.



17 [Challenging] • Target: The binomial theorem • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the coefficient of x^3y^2 in the expansion of $(x + y)^5$.

18 [Challenging] • Target: The binomial theorem • Cross-Lesson: 4.01 Lines, Triangles & Polygons

In the expansion of $(2x - 1)^4$, find the coefficient of x^2.

19 [Challenging] • Target: The normal distribution • Cross-Lesson: 5.01 Trigonometry

A data set is normally distributed with mean 100 and standard deviation 15. What percent of values lie between 70 and 130?

20 [Challenging] • Target: The normal distribution • Cross-Lesson: 6.01 Circles & Coordinate Geometry

For the same distribution (mean 100, standard deviation 15), find the approximate percentage of values greater than 115.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Permutations

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 8 members?

a. 512

b. 330

c. 342

d. 336

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22 [Foundational] • Target: Combinations

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 16 members?

a. 3360

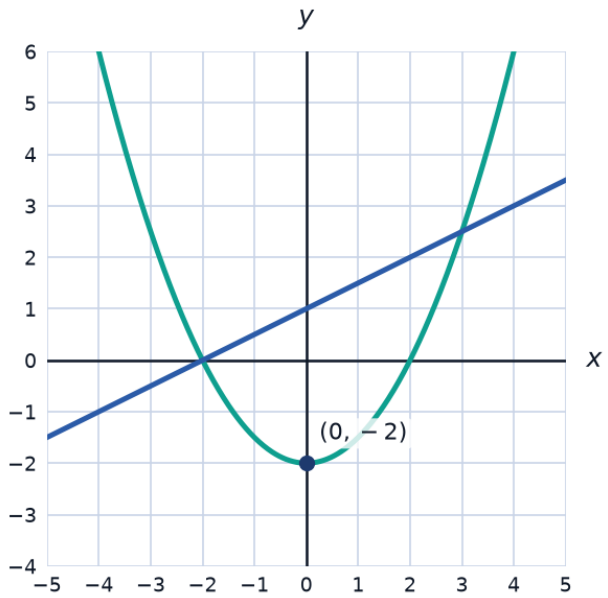
b. 4096

c. 3366

d. 3354

23 [Foundational] • Target: Probability rules

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 10 members?



a. 714

b. 726

c. 720

d. 1000

24 [Foundational] • Target: Conditional probability and independence

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 20 members?

a. 6834

b. 6840

c. 6846

d. 8000

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25 [Foundational] • Target: Measures of center and spread

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 9 members?

a. 504

b. 498

c. 729

d. 510

26 [Foundational] • Target: Data representation

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 6 members?

a. 216

b. 114

c. 120

d. 126

27 [Foundational] • Target: The binomial theorem

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 11 members?

a. 996

b. 984

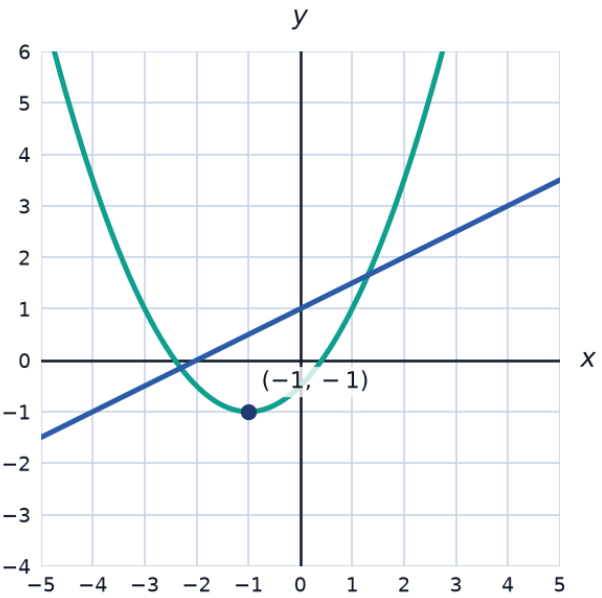
c. 1331

d. 990

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28 [Foundational] • Target: The normal distribution

How many ways can a committee of 5 be chosen from 11 people?



a. 465

b. 459

c. 462

d. 467

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: quality control sampling

How many ways can a committee of 5 be chosen from 19 people?

a. 11631

b. 27132

c. 11628

d. 11625

30 [Intermediate] • Target: Counting Principles & Probability

How many ways can a committee of 4 be chosen from 17 people?

a. 2383

b. 6188

c. 2380

d. 2377

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31 [Intermediate] • Target: Conditional Probability & Independence

How many ways can a committee of 4 be chosen from 20 people?

a. 4842

b. 4845

c. 4848

d. 15504

32 [Intermediate] • Target: Normal Distributions & Standard Scores

How many ways can a committee of 5 be chosen from 16 people?

a. 8008

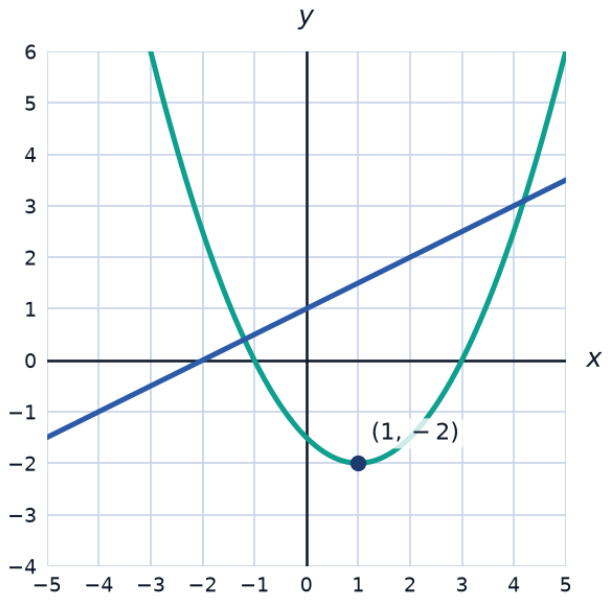
b. 4371

c. 4365

d. 4368

33 [Intermediate] • Target: Regression & Correlation

How many ways can a committee of 4 be chosen from 7 people?



a. 21

b. 32

c. 38

d. 35

34 [Intermediate] • Target: Permutations

Events C and D are independent, with $P(C) = 0.2$ and $P(D) = 0.9$. Find $P(C \cup D)$.

a. 0.97

b. 0.82

c. 0.92

d. 1.02

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35 [Intermediate] • Target: Combinations

Events C and D are independent, with $P(C) = 0.3$ and $P(D) = 0.3$. Find $P(C \cup D)$.

a. 0.56

b. 0.41

c. 0.61

d. 0.51

36 [Intermediate] • Target: Probability rules

Events C and D are independent, with $P(C) = 0.6$ and $P(D) = 0.6$. Find $P(C \cup D)$.

a. 0.94

b. 0.89

c. 0.84

d. 0.74

37 [Intermediate] • Target: Conditional probability and independence

Find the coefficient of x^7y^3 in the expansion of $(x + y)^{10}$.

a. 118

b. 122

c. 120

d. 210

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Measures of center and spread • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Find the coefficient of x^6y^3 in the expansion of $(x + y)^9$.

a. 82

b. 126

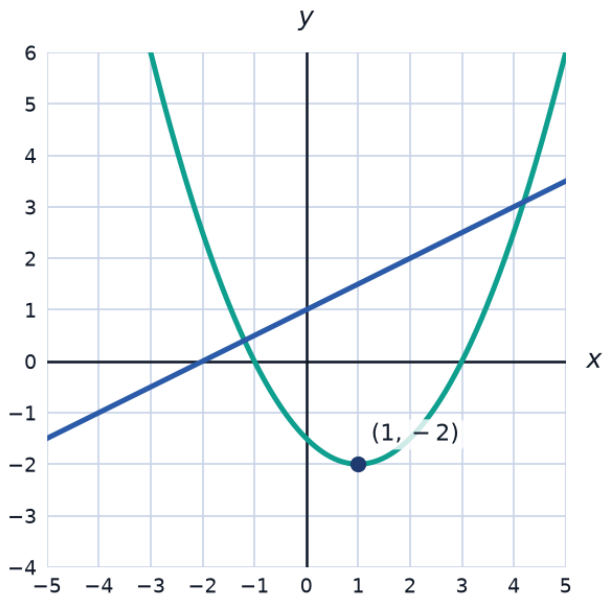
c. 86

d. 84

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39 [Challenging] • Target: Data representation • Cross-Lesson: 5.01 Trigonometry

From a group of 10 men and 8 women, how many ways can a committee of 5 be formed with exactly 3 men and 2 women?



a. 148

b. 3350

c. 3370

d. 3360

40 [Challenging] • Target: The binomial theorem • Cross-Lesson: 6.01 Circles & Coordinate Geometry

From a group of 7 men and 7 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?

a. 42

b. 441

c. 431

d. 451

41 [Challenging] • Target: Counting Principles & Probability • Cross-Lesson: 7.01 Solid Geometry & Similarity

A password consists of 3 letters followed by 2 digits. If repetition of letters is NOT allowed, how many such passwords can be created?

a. $1,560,000$

b. $1,757,600$

c. $1,404,000$

d. $2,600,000$

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42 [Challenging] • Target: Conditional Probability & Independence • Cross-Lesson: 1.01 Advanced Algebra

If $P(A) = 0.4$, $P(B \mid A) = 0.75$, and $P(B) = 0.5$, what is $P(A \mid B)$?

a. 0.60

b. 0.50

c. 0.30

d. 0.75

43 [Challenging] • Target: Conditional Probability & Independence • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In a certain school, 60% of students take French, 40% take Spanish, and 20% take both. What is the probability that a student takes French given that they take Spanish?

a. 0.50

b. 0.33

c. 0.60

d. 0.25

44 [Challenging] • Target: Normal Distributions & Standard Scores • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A dataset has mean $\mu = 100$ and variance $\sigma^2 = 25$. What score corresponds to a z-score of 1.8?

a. 109

b. 105

c. 118

d. 145

45 [Challenging] • Target: Regression & Correlation • Cross-Lesson: 4.01 Lines, Triangles & Polygons

If the correlation coefficient between two variables is $r = -0.85$, what does this indicate?

a. A strong negative linear association

b. A weak negative linear association

c. A strong positive linear association

d. No linear association

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